TeleCon connection modes

What to select before you connect

Phone setting, board, and sketch must match

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Abstract

A connection mode is the link the phone opens to the ESP32. Pick it in the application settings before you tap Connect. The sketch flashed on the board must use the same handshake. This note lists every mode the app still offers, and the camera overlay, which is not a separate mode.

Camera overlay

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1 Set the mode before you connect

The phone does not change the sketch when you change the mode. It only changes how it speaks. Choose the mode, then connect. Classic Simple, Classic Binary, and BLE Binary are all available.

- The mode stays in effect for that application until you change it. You do not reconnect just to remap a telemetry channel. You do reconnect, and you reflash, when the handshake itself changes.

2 Modes at a glance

Mode in the app	Board	Sketch
Classic + Simple	DevKit	`TeleCon_Classic_Simple`
Classic + Binary	DevKit	`TeleCon_Classic_Binary`
BLE + Binary	DevKit	`TeleCon_BLE_Binary`

The optional row **Use SoftAP camera** sits on top of the three Bluetooth modes. It does not replace them. See the last section.

3 Classic + Simple

Bluetooth Classic, text lines. This is the first mode to use.

- App: board **DevKit**, mode **Classic + Simple (text lines)**.
- Board name `ESP32-TC-RC-BT-Simple`, PIN `1234`.
- Handshake `RC:CONNECT,proto,simple`.
- Sticks, buttons, and telemetry are lines such as `RC:CTRL`, `RC:BTN`, `RC:DATA`, and `RC:PLOT`.
- Eight telemetry bytes (CH1–CH8) when the sketch sends a plot.

4 Classic + Binary

Same Classic radio as Simple, with short binary frames instead of text. Higher rate, same pins, less to read in the Serial Monitor.

- App: board **DevKit**, mode **Classic + Binary**.
- Board name `ESP32-TC-RC-BT-Binary`.
- Handshake `RC:CONNECT,proto,binary`.
- Frames start with `AA55` (sticks), `BB66` (buttons), and `CC11/22/33` (telemetry).
- Do not pair this board as BLE, and do not leave the phone on Simple.

5 BLE + Binary

Bluetooth Low Energy, Nordic UART, with the same binary frames as Classic Binary. Use it for a BLE-only module such as ESP32-C3 or ESP32-C6, or when you want the lower idle radio of BLE.

- App: board **DevKit**, mode **BLE + Binary**.
- Board name `ESP32-TC-RC-BLE-Binary`.
- Handshake `RC:CONNECT,proto,binary`.
- There is no Classic PIN. Scan for the BLE name. Do not pair it as a Classic device.

6 SoftAP camera is not a mode

Use SoftAP camera is a switch on a DevKit Bluetooth mode. The control mode stays Classic Simple, Classic Binary, or BLE Binary.

- Flash `TeleCon_CAM_SoftAP_Video` on a separate ESP32-CAM.
- Join `TeleCon-RC-CAM-Starter` / `telecon1234` for the picture only.
- Sticks stay on the Bluetooth mode you already selected.
- The phone does not open TCP port 3333 for this overlay.

One ESP32-CAM does not carry the sticks. A DevKit does Bluetooth, and a second CAM does video only.